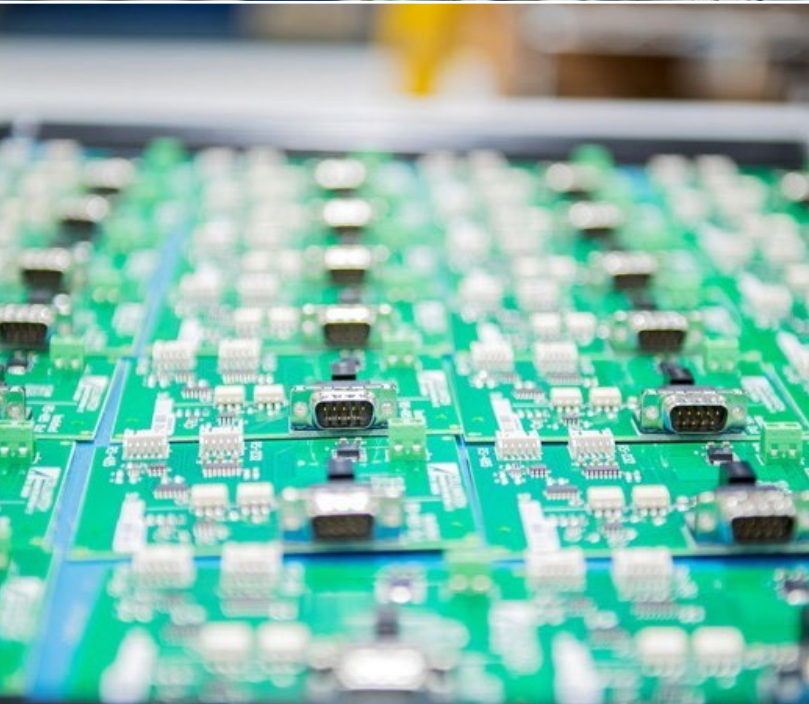


White Paper

Workload Consolidation for Industrial Automation and Robotics

Unifying Control, I/O, Visualization, and Edge AI
on a Rugged Embedded Platform

SEALEVEL®



Workload Consolidation for Industrial Automation and Robotics

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Executive Summary

Industrial automation and robotics platforms now execute more workloads at the edge. A single machine or robotic subsystem acquires sensor data, executes control logic, processes high-bandwidth camera streams, runs AI inference, supports local visualization, communicates with higher-level systems, and maintains diagnostic visibility across long deployment lifecycles.

The traditional response has been to add dedicated hardware for each new function: a controller for machine behavior, an industrial PC for HMI, a vision computer for inspection, a gateway for connectivity and security, and additional devices for data logging or analytics. That architecture remains valid where workloads require physical separation. As systems scale, however, it increases hardware nodes, cabling, power budgets, thermal load, software images, update paths, and validation effort.

Workload consolidation provides a streamlined architecture. It uses virtualization, workload partitioning, and modular compute to organize multiple operating environments and application zones on fewer rugged embedded platforms. The objective is not to collapse every function into a single software stack. The objective is to allocate CPU cores, memory, storage, I/O, and communication paths according to workload behavior while preserving isolation between functions with different timing, security, and lifecycle requirements.

For Sealevel and congatec, the proposed architecture combines three complementary layers: Relio R1 HPC+ as the rugged deployment platform; congatec COM-HPC Mini as the modular compute foundation; and aReady.VT as the virtualization technology positioned to support partitioned workloads on a Computer-on-Module (COM)-based platform, subject to validated configuration. Together, these layers establish a practical path for consolidating control, I/O, visualization, data acquisition, connectivity, and edge AI workloads. This approach is most effective when workload behavior, I/O ownership, thermal headroom, and validation boundaries support shared-platform deployment.

Consolidation reduces hardware complexity without sacrificing workload isolation.

Why Workload Consolidation Now

Industrial Automation and Robotics are Reaching a Compute Architecture Inflection Point

Automation and robotics systems are moving from fixed-function control toward software-defined edge architectures. The shift is driven by AI-enabled inspection, predictive maintenance, high-bandwidth sensing, remote diagnostics, adaptive robotics, and tighter integration between operational technology and higher-level data systems.

Workload growth matters because these functions are now interdependent.

- Inspection outputs feed machine sequencing.
- Sensor streams serve both control loops and analytics services.
- Diagnostic workloads require visibility into I/O activity, network status, and application health.
- Visualization layers combine AI inference results, control state, and connectivity data into operator-facing views.

As these dependencies increase, distributed hardware architectures become harder to manage. Each added device introduces another power input, network interface, enclosure or mounting requirement, software image, security considerations, and lifecycle path. Troubleshooting spans more devices and integration points. Validation expands because changes in one device alter communication behavior across the system.

Workload consolidation addresses this pressure by reducing unnecessary hardware fragmentation under defined engineering conditions. Related workloads operate on a shared embedded platform while remaining partitioned by function, operating environment, resource requirement, and criticality. The practical benefit is not simply a lower device count; the engineering outcome is a more coherent compute architecture with defined workload boundaries, controlled data paths, and a clearer basis for lifecycle planning.

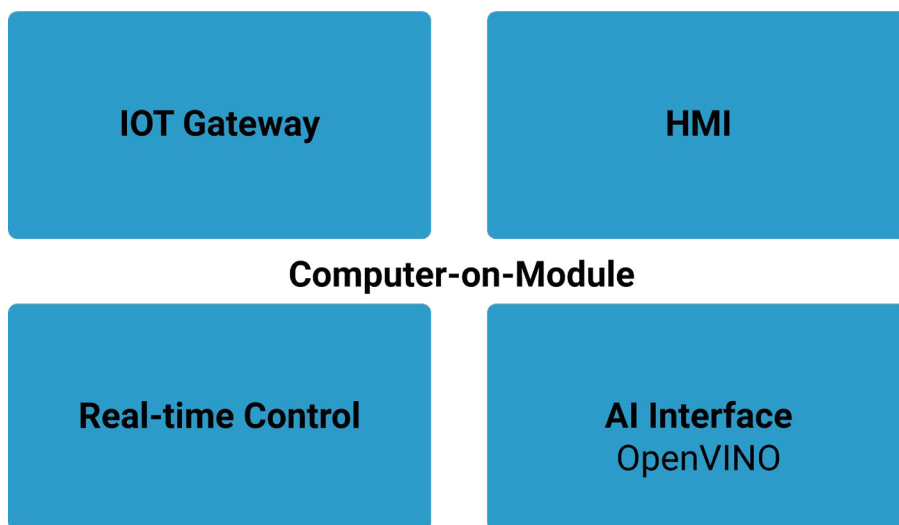


Figure 1: Traditional distributed architecture vs. workload-consolidated architecture.

The Workload Collision

Why More Compute Does Not Always Mean More Boxes

The core engineering challenge is workload collision: multiple software domains with different timing, compute, I/O, operating system, and validation requirements competing for resources on the same platform or across an increasingly complex distributed architecture.

Machine vision workloads are bursty and compute-intensive. Control workloads require deterministic or time-sensitive execution when they manage motion, sequencing, or deterministic I/O. Visualization workloads depend on Windows or Linux application frameworks and responsive graphics. Data services are network- and storage-intensive but typically less timing-sensitive. Connectivity workloads run continuously in the background and require defined security boundaries.

Balancing these workloads becomes unwieldy when they interact. AI inference results trigger downstream control decisions. HMI displays consume operational data from control and diagnostics zones. Data logging captures machine state and inspection output. Mobile robotics platforms coordinate perception, localization, path planning, motion control, telemetry, and service diagnostics inside constrained power and mechanical envelopes.

Before workloads are consolidated, the architecture must answer several questions:

- 1. Which functions require deterministic behavior?
- 2. Which workloads need direct I/O or PCIe ownership?
- 3. Which zones share networking or storage?
- 4. Which operating systems remain in the architecture?
- 5. How do VMs exchange data without undermining isolation?
- 6. What headroom is required for peak AI, vision, or communications load?

Workload Zone	Typical Functions	Primary Architecture Concern	Consolidation Implication
Control and I/O	Motion control, sequencing, deterministic I/O, industrial communications	Predictable execution, I/O ownership, interrupt behavior	Assign dedicated resources where timing or direct device access is required
Edge AI and machine vision	Object detection, defect inspection, anomaly detection, classification	Compute bursts, accelerator access, camera bandwidth	Isolate from control zones; define data handoff to control or data services
Visualization and HMI	Operator dashboards, alarms, configuration tools, supervisory interfaces	Application framework, UI responsiveness, update cycles	Separate from timing-sensitive workloads; support controlled access to operational data
Data and connectivity	Logging, diagnostics, predictive maintenance, cloud/edge communication	Network/storage throughput, security boundary, continuous operation	Partition communication paths and define telemetry access without broad device sharing

For robotics platforms, the same challenge appears in compact and mobile form. Perception, localization, path planning, motion control, telemetry, and diagnostics coordinate continuously while maintaining responsiveness. Reliable operation depends on workload organization, timing analysis, I/O planning, and lifecycle management — not on processor performance alone.

What Workload Consolidation Means in Engineering Practice

Workload consolidation is the process of combining software functions that traditionally ran on separate hardware systems into a coordinated computing architecture. In this paper, workload consolidation is the primary term. **System consolidation** describes the resulting architecture, and **hardware consolidation** describes the physical outcome of reducing dedicated devices when the application allows it.

Virtualization makes this model practical by creating independent software-defined environments called virtual machines (VMs). Each VM runs its own operating system, application stack, resource allocation, and update path. To the application, the VM behaves like a dedicated computing environment; to the system architect, it becomes one partitioned workload zone within a larger platform architecture.

This distinction matters in industrial systems. Workload consolidation does not merge all software into one monolithic image. A consolidated architecture preserves separation between AI inference, control, HMI, diagnostics, and connectivity functions while allowing selected data exchange between them. The result is a modular software architecture with functional boundaries that support validation, maintenance, and lifecycle planning.

Term	Technical Meaning	Use in This Paper
Workload consolidation	Hosting multiple application workloads or operating environments on a coordinated platform architecture	Primary term for the engineering strategy
System consolidation	Combining functions that previously required separate systems into a single system-level architecture	Supporting term for the broader architecture
Hardware consolidation	Reducing the number of physical computers, controllers, gateways, or vision systems	Supporting term for plain-English benefits such as lower hardware count and less cabling
Workload partitioning	Assigning resources and isolation boundaries to each workload zone	Core technical mechanism enabled by virtualization/hypervisor technology

Technical Note: Workload consolidation is not just running multiple applications on one computer. It is an architecture for separating, assigning, and coordinating workloads that have different timing, I/O, operating system, and lifecycle requirements.

Enabling Technology 1: Virtualization, Hypervisors, and aReady.VT

A hypervisor is the software layer that creates and manages virtual machines. In an embedded workload-consolidation architecture, the hypervisor also influences how processor cores, memory, I/O devices, timers, storage, and communication mechanisms are allocated across workload zones.

For industrial applications, the relevant capability is controlled separation, not merely running multiple operating systems on one processor. A hypervisor partitions hardware resources so that control, AI, HMI, and data services zones operate with defined resource boundaries. Where coordination is required, the architecture uses explicit inter-VM communication mechanisms instead of uncontrolled shared dependencies.

congatec describes hypervisors as enabling communication and coordination between VMs while isolating them so they do not compromise each other's operation. The congatec source diagrams used to inform this paper's visual plan illustrate concepts such as time and space isolation, hardware partitioning, core allocation, I/O/RAM/PCIe assignment, and inter-system communication through shared memory, virtual networking, and event systems.

This forms the technical foundation for mixed automation workloads. In a validated design, an AI inspection VM publishes inspection results to a control VM. A visualization VM reads selected state information from control or data services. A diagnostics VM collects telemetry without owning the control I/O path. The design intent is controlled communication with preserved workload boundaries.

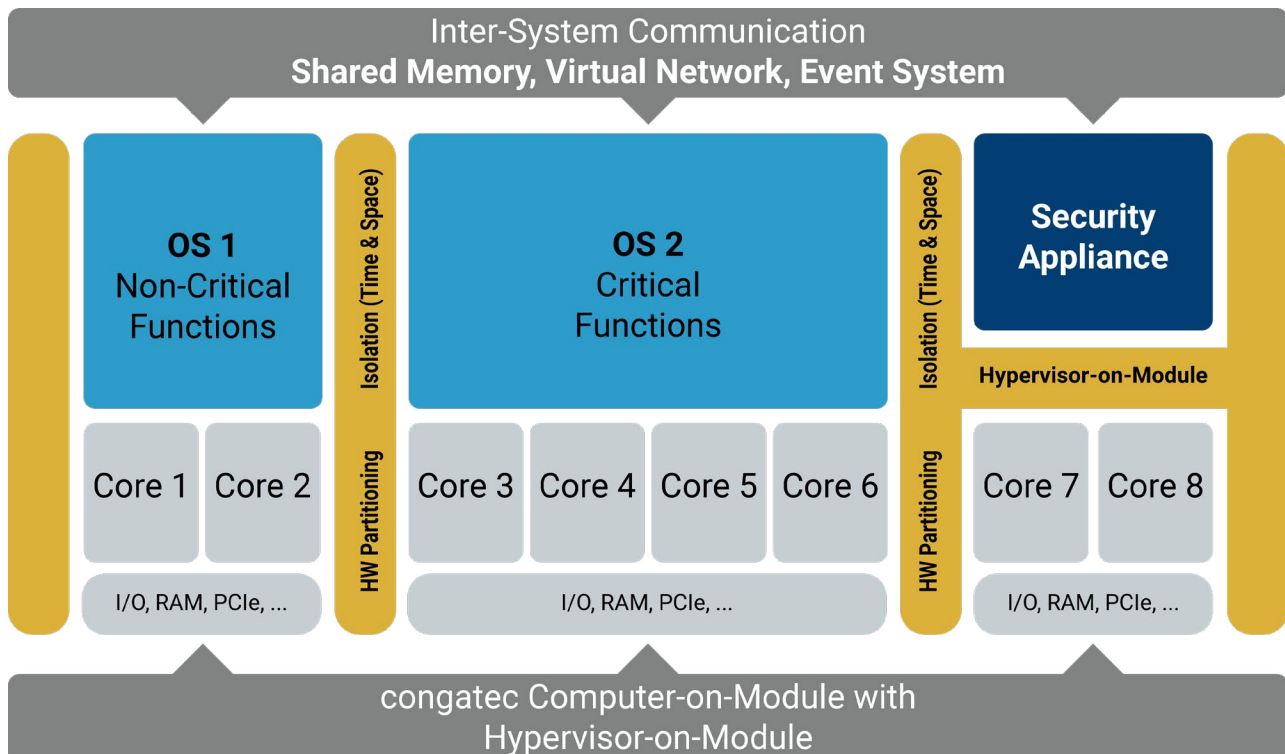


Figure 2: Hypervisor-based workload partitioning

Within the proposed Sealevel + congatec architecture, aReady.VT is positioned as the virtualization technology that supports workload partitioning on the COM-based platform. Final implementation language must be validated by congatec and Sealevel against the specific Relio R1 HPC+ configuration, selected COM-HPC Mini module, licensed software stack with congatec hypervisor conga-zones as part of aReady.VT, supported operating systems, and intended workload mix.

Enabling Technology 2: COM-HPC Mini and Modular Compute

Workload consolidation requires a structural compute foundation that scales with the application and remains supportable over the life of the machine. In industrial and robotics deployments, the compute architecture must support current workload requirements while preserving a path for processor, memory, I/O, and software evolution.

COM architectures are well suited to this problem because they separate the compute engine from the application-specific system design. congatec describes COMs as application-ready supercomponents that integrate core PC components such as CPU, GPU, memory, and peripheral I/O into a standardized module. The carrier or system design then focuses on mechanical integration, I/O, power, thermal design, and application-specific form, function, and interfaces.

That separation is directly relevant to workload consolidation. AI inference, visualization, communications, and data services often evolve faster than the machine frame, enclosure, mounting approach, or field wiring. A COM-based architecture gives the system designer a defined migration path for compute capability without redesigning every application-specific element of the platform.

COM-HPC Mini provides a compact modular compute foundation for high-performance embedded systems. For the final platform language, module-level claims should align with the selected congatec module and validated Sealevel system configuration, including processor family, memory configuration, high-speed interfaces, thermal design range, graphics/AI capabilities, and lifecycle expectations.

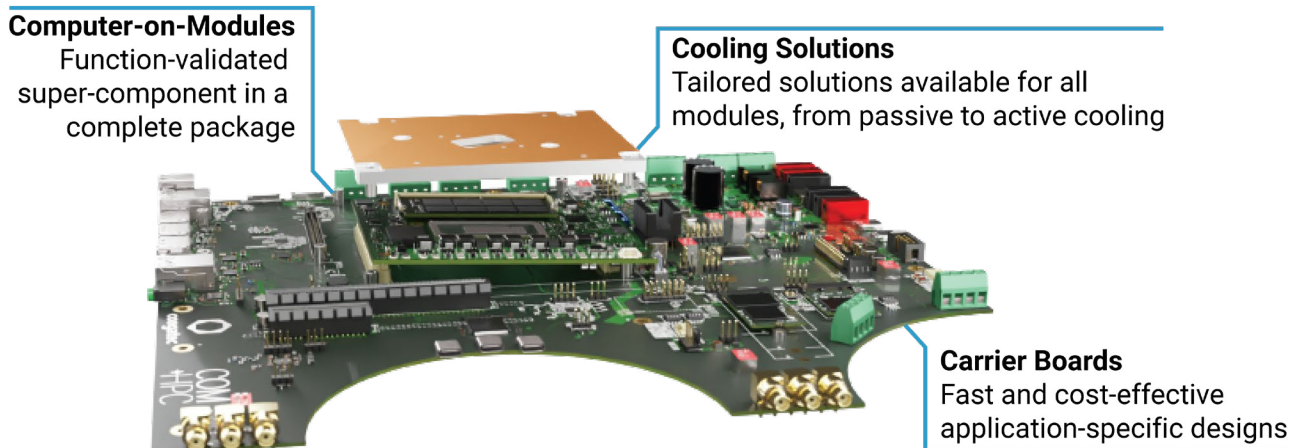


Figure 3: COM-HPC Mini as modular compute foundation

Enabling Technology 3: Relio R1 HPC+ as the Deployment Platform

Virtualization and modular compute define the architecture. Relio R1 HPC+ is the rugged edge deployment platform that places that architecture near the machine, sensor, camera, robot, or control asset.

This platform role is critical. A consolidated workload architecture creates operational value only when the embedded computer installs where workloads execute: in or near a control cabinet, machine enclosure, mobile platform, inspection cell, test stand, or industrial edge node.

Sealevel's Relio R1 HPC+ uses a COM-HPC architecture with Intel Core i5 and Intel Core i7 industrial processor options. Configurations include the congatec HPC/mRLP-i5-1345URE-16G Core i5 CPU and congatec HPC/mRLP-i7-1365URE-32G Core i7 CPU, with up to 32 GB DDR5 memory. These Intel-based configurations provide the compute foundation for vision, high-bandwidth data handling, and mixed edge workloads.

The platform's connectivity and expansion options support the workload-consolidation use case. The Relio R1 HPC+ features five 2.5 GbE ports, optional Wi-Fi 6E, two SeaLATCH locking USB 3.1 Type A ports, two SeaLATCH locking USB 3.1 Gen 2 Type C ports with DisplayPort Alt Mode, two SeaLATCH locking USB 2.0 Type A ports, one dedicated DisplayPort, and an isolated serial console through USB Type C. Expansion options include CFexpress storage, E1.S drive support, M.2 M-key NVMe storage, a Unigen Hailo-8 Biscotti E1.S AI accelerator option and other NVMe-based AI accelerator cards, and CAN bus expansion depending on configuration.

The industrial deployment specifications reinforce Sealevel's platform-level role with a fanless, solid-state anodized aluminum enclosure with intentional thermal management, an operating temperature range of -40 C to +50 C, storage temperature range of -40 C to +85 C, and 10-90% non-condensing relative humidity. The Relio R1 HPC+ includes locking SeaLATCH USB connectors, 9-48 VDC input power, and an 8 in. x 5.75 in. x 2.4 in. compact footprint.

The software environment includes Windows 10 IoT Enterprise, Windows 11 IoT Enterprise, Linux RHEL 8.7 or later, and Linux Ubuntu 20.04 or later. These operating system options support separate application environments for HMI, AI/vision, data services, and edge software. VM count, conga.zones configuration, RTOS support, I/O assignment, and inter-VM communication mechanisms remain part of the validated reference architecture review.

Rugged edge computing brings high-performance, consolidated workloads closer to the machines, sensors, and systems they serve.

Reference Architecture

Consolidating AI, Control, I/O, and Visualization

The reference architecture organizes workloads into functional zones rather than individual applications. This keeps the architecture general enough for multiple automation and robotics use cases while remaining specific enough for technical readers to understand how partitioning, I/O ownership, and data exchange are planned.

A conceptual configuration includes four workload zones:

1. Edge AI and Machine Vision
2. Control and I/O
3. Visualization and HMI
4. Data and Connectivity

Each zone maps to an operating environment, resource allocation, I/O dependency, and inter-zone communication requirement. The architecture remains conceptual until Sealevel and Congatec validate the exact hardware and software stack.

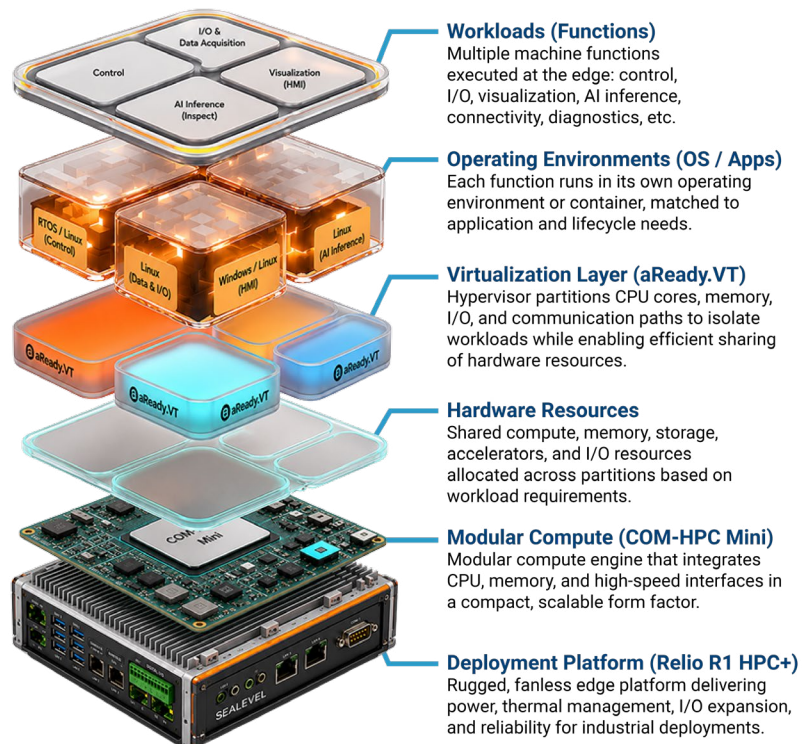


Figure 4: SeaLevel and Congatec Workload Consolidation Stack

Zone	Representative Workloads	Resource / I/O Considerations	Communication Requirements
Edge AI and Machine Vision	Object detection, image classification, defect inspection, sensor fusion	Camera bandwidth, GPU/CPU/accelerator availability, memory bandwidth, storage for images/results	Publishes inference results to control, HMI, and/or data services.
Control and I/O	Machine sequencing, motion coordination, industrial communications, deterministic I/O	Dedicated cores where required, direct I/O or PCIe ownership, predictable interrupt handling	Consumes AI or sensor status when validated; publishes machine state to HMI/diagnostics.
Visualization and HMI	Dashboards, alarms, configuration tools, service screens	Graphics/UI responsiveness, access to selected operational data, update workflow	Reads state and results; writes configuration only through controlled, validated interfaces.
Data and Connectivity	Logging, remote diagnostics, predictive maintenance, cloud/edge communication	Network throughput, storage endurance, cybersecurity boundary, service access	Aggregates data from other zones and exports through defined channels

Use Case: Automated Inspection and Control Cell

An automated inspection and control cell is a strong example because it combines high-bandwidth data, time-sensitive machine behavior, operator visibility, and quality reporting. A vision system must acquire images, run inspection logic, classify pass/fail or defect conditions, and communicate results. The control system must coordinate conveyor movement, sensors, actuators, and I/O. Operators need HMI visibility into state, alarms, and inspection output. Data services handle quality records, diagnostics, predictive maintenance inputs, and higher-level system communication.

Traditional Node	Consolidated Zone	Engineering Consideration
Vision PC	Edge AI / Machine Vision VM	Camera bandwidth, model runtime, accelerator availability, result handoff
Controller or industrial PC	Control and I/O VM	I/O ownership, deterministic behavior, interrupt handling, safety-adjacent boundaries
HMI system	Visualization / HMI VM	Operator responsiveness, application framework, update path
Gateway or logging device	Data and Connectivity VM	Network segmentation, storage throughput, remote diagnostics

In a distributed architecture, each function often runs on a dedicated hardware node. That approach simplifies physical separation but increases hardware count and integration paths. In a workload-consolidated architecture, these functions operate as separate workload zones on a rugged embedded platform.

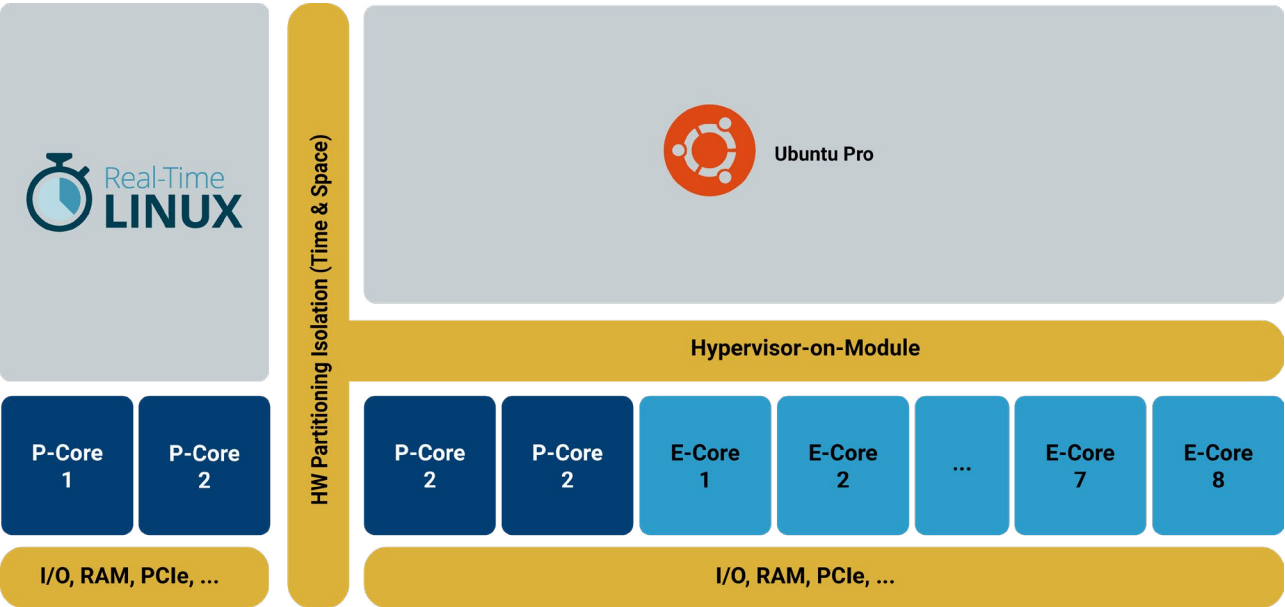


Figure 5: Automated Inspection Cell Workload Map

The control zone retains assigned resources and I/O access. The AI/vision zone receives camera data and performs inference or inspection. The HMI zone accesses selected state and results. The data services zone logs events, stores inspection metadata, or forwards information to higher-level systems.

The engineering value comes from explicit workload boundaries. The control path, inspection path, visualization path, and data path are designed, validated, and maintained as partitioned zones with controlled communication between them.

Use Case: Robotics and Mobile Automation

Robotics and mobile automation platforms compress multiple workload types into a constrained mechanical, power, and thermal envelope. Perception, localization, path planning, motion control, telemetry, diagnostics, and operator/service interfaces must operate concurrently while exchanging data at defined points in the control and autonomy stack.

[congatec's AMR white paper](#) provides strong source support for this use case. It describes robotics systems built around sensors such as lidar, 2D/3D cameras, ultrasonic sensors, infrared sensors, and encoders; AI algorithms for object recognition and obstacle detection; SLAM/localization; path planning; collision avoidance; and motion control. It also emphasizes timing synchronization, resource allocation, modular software organization, and the risk of non-critical functions interfering with critical robotics functions.

A workload-consolidated architecture maps these functions into VM zones when the workload behavior supports shared-platform deployment. Perception and AI workloads are separated from motion-control workloads. Localization and path planning exchange data through defined communication mechanisms. Telemetry and diagnostics operate without owning the control I/O path. Service interfaces follow an update workflow that limits impact on the control or perception zones.

Robotics Function	Possible Worked Zone	Primary Design Consideration
Sensor acquisition and perception	Edge AI / Machine Vision VM	High-bandwidth sensor input, model runtime, memory bandwidth
SLAM / localization	Edge AI or Data Processing	Continuous sensor fusion, map/state update rate, compute headroom
Path planning	Control-adjacent compute zone	Dependency on perception/localization, responsiveness to dynamic obstacles
Motion control	Control and I/O	Predictable execution, actuator I/O, direct device access where required
Telemetry and diagnostics	Data and Connectivity	Network availability, service access, system health data, cybersecurity boundary

Engineering Benefits of Workload Consolidation

The strongest case for workload consolidation is engineering control over complexity. When applied to appropriate workloads, the architecture reduces hardware nodes, simplifies interconnects, improves resource utilization, and establishes clearer boundaries for software maintenance and lifecycle planning.

Reduced hardware count is the most visible benefit, but it should not be the only message. For engineering teams, the more important benefits are resource allocation, workload isolation, explicit communication paths, and modular update boundaries. These factors determine whether the architecture is maintainable after deployment.

Benefit	Technical Mechanism	Reader Value
Reduced system complexity	Fewer dedicated devices and fewer hardware interconnects	Less wiring, fewer integration points, simplified enclosure and service planning
Improved resource utilization	CPU, memory, I/O, and storage allocated by workload need	Modern multicore compute supports multiple zones without dedicating one hardware node per function when timing, I/O, and thermal constraints are satisfied
Workload isolation	VM boundaries and hardware partitioning	Properly partitioned non-critical workloads remain isolated from timing-sensitive functions under validated configuration
Controlled data exchange	Shared memory, virtual networking, event systems, or validated interfaces, resilient communications between VMs, reduces the attack surface, improves security	Workloads coordinate through defined interfaces instead of uncontrolled dependencies
Lifecycle flexibility	COM-based compute and modular software zones	Future compute migration and software updates follow defined system boundaries

Engineering Note: Validate the architecture under workload stress, not only under nominal conditions. AI inference bursts, camera bandwidth, network traffic, storage activity, and HMI updates expose resource conflicts that are not visible in a static block diagram.

Why Consolidate?



Fewer Hardware Nodes



Fewer Interconnects



Better Resource Utilization



Stronger Isolation and Security



Simpler Updates and Lifecycle

Design Considerations for Consolidated Automation Platforms

Workload consolidation requires architectural discipline. The system should be defined around workload behavior, not simply around peak processor specifications. A processor with sufficient aggregate performance still fails the application if I/O ownership, latency, memory bandwidth, thermal headroom, or validation boundaries are not planned correctly.

Design Activity	Technical Questions	Output Needed
Workload mapping	Which workloads are deterministic, compute-intensive, I/O-bound, security-sensitive, or update-sensitive?	Workload inventory and criticality classification.
Resource allocation	Which zones need dedicated cores, memory, storage, network interfaces, or accelerator access?	Resource allocation plan by VM/zone.
Operating system strategy	Which workloads require Windows, Linux, RTOS, real-time Linux, or other supported environments?	Validated OS/VM matrix.
I/O partitioning	Which devices require direct ownership? Which devices are virtualized or shared?	I/O ownership and device assignment map.
Inter-VM communication	What data must cross zones, how often, and with what latency or synchronization requirement?	Communication interface design.
Thermal and power planning	What is the combined load under peak AI, vision, network, storage, and control activity?	Thermal/power budget with headroom.
Lifecycle validation	How will updates, module migrations, OS patches, and workload additions be validated?	Lifecycle and regression test plan.

This framework also defines Sealevel's consultative role. Configuration planning identifies workload behavior early. Integration planning evaluates I/O, networking, legacy equipment, and data paths. Lifecycle planning addresses component availability, compute migration, software support windows, and service requirements before the architecture is frozen.

How Do You Consolidate?



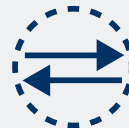
Map workloads and classify by behavior, criticality, and update sensitivity.



Allocate compute, memory, storage, network, and accelerators.



Partition workloads by OS/VM environment, I/O ownership, and security requirements.



Integrate I/O, networks, and data paths across workload zones.



Validate performance, thermal headroom, isolation, and lifecycle processes.

Why Sealevel + congatec

congatec and Sealevel address different layers of the workload consolidation problem.

congatec provides the modular compute and virtualization foundation. Its COM-based approach aReady.COM supports scalable embedded processing, while aReady.VT is positioned to consolidate functions that previously required multiple systems onto one COM-based platform. congatec's aReady.COM application-ready modules also support the use of preconfigured operating systems, Hypervisor-on-Module conga-zones, and application-ready software building blocks, subject to product and configuration validation.

Sealevel provides the rugged embedded system layer and application integration perspective. Relio R1 HPC+ translates modular compute and virtualization into a deployable industrial system with Intel Core i5/i7 processor options, high-throughput Ethernet, locking USB connectivity, display support, storage and AI-accelerator expansion paths, rugged mechanical design, thermal management, and lifecycle-oriented platform support, subject to final datasheet validation.

Together, the partnership presents a practical architecture for consolidating more automation workloads onto fewer rugged systems. The strongest message is technical and customer-focused: Sealevel and congatec help engineering teams evaluate where workload consolidation is appropriate, partition workloads by behavior and resource needs, and deploy a modular embedded platform that supports long-lifecycle automation and robotics systems.

Looking Ahead

Industrial systems will continue to add software-defined functions: AI-enabled inspection, predictive maintenance, cybersecurity services, remote diagnostics, sensor fusion, fleet coordination, and more advanced operator interfaces. Each new function introduces compute, data, security, validation, and lifecycle implications.

Engineering teams need a method for adding those capabilities without increasing hardware count, recabling systems, fragmenting software maintenance, and restarting validation work. Workload consolidation provides a framework for organizing new functions as defined workload zones rather than as separate hardware projects by default.

Conclusion and Key Takeaways

Workload consolidation is relevant because automation and robotics systems are running more software-defined workloads closer to the machine. AI inference, machine vision, control, I/O, visualization, connectivity, and diagnostics increasingly need to operate together, but they do not all behave the same way.

Key Takeaway	What It Means for Engineering Teams
Workload consolidation is an architecture strategy.	Plan around workload behavior, not only hardware reduction.
Virtualization enables isolation and coordination.	VMs and hardware partitioning support mixed workloads with defined communication paths.
COM-HPC Mini supports modular compute.	COM-based compute separates technology migration from system-level I/O and mechanics.
Relio R1 HPC+ is the deployment layer.	Sealevel's value is rugged deployment, I/O integration, and lifecycle support.
Validation remains essential.	Verify OS support, I/O ownership, thermal headroom, licensing, and workload behavior.

Workload consolidation is not about putting every application on one system. It is about identifying where consolidation creates measurable architectural value, partitioning workloads with the right isolation and communication mechanisms, and deploying the resulting architecture on hardware designed for industrial environments.



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